

PEGASUS

Final Report

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Abstract

To determine the advantages and practicality of a petroleum/gasoline substitute system for an internal combustion engine, an onboard fuel reformer was constructed by the Fall 2012 PEGASUS senior design project team of Old Dominion University. The objectives of the project, led by faculty member Dr. Stephen Knisley, and consisting of students David Rowe, Marcus Updyke, and Brandon Pape, were to research, design and construct a petroleum/gasoline substitute system whose operation could be fully explained and demonstrated. The system was designed to allow the ability to operate on a wide variety of alternate fuels as well as increase the efficiency of those fuels as compared to a standard internal combustion engine. Additionally, the system was designed such that it could be easily dismantled and equipped with testing equipment for future research teams working on the PEGASUS project. Following research on varying applicable systems, the petroleum/gasoline substitute system that was chosen to be designed and built is known as an onboard fuel reformer. The general operation of the fuel reformer consisted of two main phases to create a hydrogen rich synthesis gas, or syngas, that was used to power the lawnmower engine. In the first phase, the engine exhaust combined with the alternate fuel and oxygen. The vaporized alternate fuel then entered the reformer of the second stage where the syngas was developed and fed to the engine manifold. The project began August 27, 2012 with the construction of the reformer and initial testing completed by November 16, 2012. The total cost of the reformer was \$272.36 with an additional \$156.45 for a lawnmower as a test device. The finished design was fitted to a standard 4.5 ft-lbf Briggs and Stratton engine mounted to a push lawnmower. The remaining project time up to December 3, 2012 was used as the testing phase where the engine, with the reformer, ran solely on gasoline, ethyl alcohol, methyl alcohol, and E85. The engine was also able to run after various mixes of all fuels were ran through the reformer, most notably a diesel mix. However, to fully assess the system a second testing PEGASUS team is expected to evaluate the reformer operation using data to further verify the efficiency gains and modify the design as necessary. The results from the design phase of the PEGASUS team show that the reformer could be constructed relatively inexpensively, fitted to an off the shelf engine, and operated on various non-petroleum fuels.

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Introduction:

Old Dominion University looks to address the dilemma of developing an internal combustion engine capable of running non-petroleum based fuels through a research project named P.E.G.A.S.U.S. (petroleum/gasoline substitute). Throughout recent history there has been an increasing reliance upon the various types of known fuels, with hydrocarbon based fossil fuels providing the bulk of our energy needs. These fuels are relied upon for power generation, heating, and the operation of numerous mechanical devices. With farm equipment, factories, and motorized vehicles all requiring fuel for their use, our economy and infrastructure heavily rely upon their use. The economic and cultural significance of fuel, as well as the impact that it has on society, cannot therefore be understated [6]. In more recent years, there has been a push for alternative and renewable forms of energy such as solar power, ocean energy, hydroelectric power, onshore and offshore wind power. This push has been the result of public concern over the future availability of fossil fuels, as well as the impact that fossil fuels have on the environment. Unfortunately, these renewable energy source have not yet become as economically viable as petroleum fuel sources and will continue to remain a small percentage of energy production in the near future [11]. However, with additional research and innovative engineering, fuel sources that were once economically infeasible and impractical can become applicable options. The objective of this project, led by faculty member Dr. Knisley, and consisting of students David Rowe, Marcus Updyke, and Brandon Pape, was to research, design, and construct a petroleum/gasoline substitute system that could be fitted to a lawnmower for demonstration. A detailed description of the operation of the reformer was also required as part of the project guidelines. This project hoped to create a synthetic gas production method to that could be used to power an engine using non-petroleum based fuels as well as increase the fuel efficiency of both petroleum and non-petroleum fuels. The petroleum/gasoline substitute system that was chosen to be designed and built is known as an on-board fuel reformer. A secondary team is anticipated to perform tests to determine the level of enhancements on efficiency and improvements on the original design.

Background:

Overview

With the increasing push to find clean and renewable sources of fuel, petroleum substitutes and new methods to raise fuel efficiency have been progressively expanding in academic research and literature. This push for new sources of energy is greatly influenced by the growing need for energy; the United States alone uses 18,835,000 barrels of petroleum per day, while 80,000,000 barrels are being used worldwide [11]. As the world of transportation is a significant consumer of this fuel, improvements in the efficiency of the internal combustion engine can create a significant impact [3]. A portion of this research has been focused on the process of reformation to create a hydrogen syngas. The onboard fuel reformer built and installed on a basic, single cylinder lawnmower engine for the PEGASUS project makes use of this previous research. Prior to construction, this research was reviewed to ensure that the fundamentals of the reformer operation could be explained in detail by the project group. Although the ability to break down and define the reformer operation was a requirement of the project, it also helped the group interpret the results of the testing process. The two main topics of the PEGASUS operation are explained in detail with a description of the reformation process and the hydrogen fuel source as well as advanced uses of this technology currently in use.

Hydrogen

A colorless, odorless, and highly combustible gas at standard atmospheric pressures, hydrogen has very important applications in the internal combustion engine field and has great potential as the fuel of the future. The fuel has numerous sources and can be found in all fossil fuels, many renewable resources, and in its most stable form, water. With over three times the BTU's of gasoline, hydrogen is a chemically rich fuel that could further the use of internal combustion engines as a complete substitute or as a mixture with normal fuels [2]. The burning of hydrogen is 100% clean, with the only byproduct being water. "It has been the fuel used to provide electricity for the space shuttle for the last two decades via on-board fuel cells that combine hydrogen and oxygen to generate electricity; the exhaust from the fuel cell – pure water – is used by the crew as drinking water" [10]. Hydrogen can either be used in engine applications as the sole fuel, or it can be used as a hybrid with another fuel. When used as a hybrid, engine

efficiency is significantly increased with emissions significantly lowered. The reason for this is due to hydrogen's high combustibility; it combusts fractions of a second before the other fuel, providing more pressure on the piston and assisting in a more complete burn of the fuel, leaving less pollutants behind to exit through the exhaust. Honda has already begun to develop this technology and is currently in the process of releasing the Honda FCX Clarity to the United States.

However, the main question to consider now is, from where can hydrogen be harvested?

Hydrogen is the most abundant element in the universe and is readily found on Earth.

Unfortunately, it is not readily found in its diatomic or monatomic form, only in molecular form with other elements forming water, alcohols and organic compounds [2]. While there are many different methods of hydrocracking molecules to break the intermolecular bonds resulting in released hydrogen, the problem has been to develop the most economic and efficient manner to extract the hydrogen. One option is electrolysis, where a voltage is applied to water and this energy breaks the bonds releasing H_2 and O_2 . While this method does work well, using conventional methods, 100 kWh of electricity is needed to produce 1 pound of hydrogen, and only 22 kWh of usable energy is available [10]. Therefore, this method is neither efficient, nor economical. If one could produce this hydrogen using solar power, wind power, or another renewable energy, this method could be efficient.

Another option is the steam reformation process in which steam is reacted with vaporized fuel at high temperature producing a hydrogen syngas. A version of steam reformation where waste exhaust gases are used in the reformation process is known as exhaust gas reformation. As an endothermic reaction, efficiency can be increased if engine exhaust is recycled to vaporize the fuel and water to be reacted [3]. Therefore, steam reformation is both efficient and economical, being the most popular method for extracting hydrogen in the present day.

Steam Reformation

The process of steam reformation is not a new technology, but improved methods are becoming more efficient and more practical for small engine use. The process of steam reformation was first used in the 1870's with the development of Town Gas, which was produced from coal and biomass. Byproducts of this process were coke, coal tar, sulfur, and ammonia, all of which are useful products. This fuel was supplied to residents of the city through a series of municipally owned piping systems and was used for cooking, heating, and the lighting of homes. This method was later replaced when gas began to be reformed from oil [12].

The process of steam reformation can be applied to an internal combustion engine considering their inefficiency. In a standard internal combustion engine, 30%-40% of the fuel's energy is converted to usable work, the other 60%-70% is lost through heat in the exhaust and radiation [13]. This 60%-70% loss of energy is where the steam (exhaust) reformation is able to obtain its energy to complete this process. Heat from the exhaust is recycled and used to vaporize the fuel, separate the hydrogen to make a clean burning and more efficient syngas. Fundamentally, the process works as energy from the exhaust is transferred to the gasified fuel in the form of heat. With enough energy, the hydrogen and other molecules are separated into a synthesis gas or syngas. Because steam reforming is an endothermic reaction, the burning value of the fuel is increased, resulting in a more efficient fuel. This process works for a multitude of non-petroleum fuels, along with all the common hydrocarbon petroleum products [2].

For the application of steam reformation for a small internal combustion engine, methanol presents itself as an excellent and viable fuel source. This simple alcohol is both easily obtainable and can be manufactured at anyone's home who may wish to do so. The equation for the process of the reformation is $\text{CH}_3\text{OH} + \text{H}_2\text{O} \Rightarrow 3 \text{H}_2 + \text{CO}_2$ (Methanol + Steam $\Rightarrow$ Hydrogen + Carbon Dioxide). It is apparent from this equation that the reformation process, given enough energy from the exhaust gas, will produce 3 mols of H_2 for each mol of CO_2 . The gas also contains some residual methanol and a small amount of carbon monoxide (CO),

typically up to 1 vol.%.” [13]. Refer to Figure 4 in the figures section for a table showing both the reformation and combustion of methane and its’ syngas respectively. Importantly, methanol has a very high molar fraction of hydrogen compared to many other fuels making it particularly attractive for reformation [5]. Regardless of the fuel, however, the process is fundamentally the same with the differences found in the chemical makeup of the syngas produced, the amount of energy required to separate the hydrogen and therefore the potential increase in efficiency. A summary of several basic methanol reformation equations can be found in Fig 7.

More advanced studies and reformers are also being investing at various institutions. MIT has been researching and developing a compact plasma steam reformer they call the “Plasmatron” since the late 90’s, and since then have implemented this system city busses with high success. MIT claims that this system could be the answer to the ever-growing international oil crisis. “If widespread use of plasmatron hydrogen-enhanced gasoline engines could eventually increase the average efficiency of cars and other light-duty vehicles by 20 percent, the amount of gasoline that could be saved would be around 25 billion gallons a year, That corresponds to around 70 percent of the oil that is currently imported by the United States from the Middle East [4].” Although the reformer currently being designed by the PEGASUS team is much more basic in structure, the concept for both devices is the same.

Research and Applications

Current research of fuel reforming has branched into several applications. These applications include fueling fuel cells, producing large scale hydrogen, reformers in cars for fuel efficiency and eliminating emissions, and producing bio-fuels. Steam reformation is the most widely used method for large scale hydrogen production and for producing hydrogen to be stored in fuel cells. Large plants have been developed such as the ATR reactor system by Haldor Topsoe in Fig 1 to produce hydrogen through autothermal reforming [9].

Current research for automotive application is centered on fuel efficiency and lower emissions. One company leading research is Toyota. Toyota is making a strong effort to advance its hydrogen power technology. The hydrogen fuel-cell cars the company offers give car buyers

a competitive alternative to gasoline-powered vehicles. Toyota's fuel-cell hybrid vehicles use an on-board methanol reformer and hydrogen-absorbing alloy that allows for the storage of pure hydrogen as seen in Fig 2 [8]. MIT has lead the research in the field of plasma reforming, where MIT research groups have been able to use electrically produced plasma to catalyst the reforming process called a Plasmatron. The Plasmatron is an onboard "oil reformer" that converts a variety of fuels into high-quality, hydrogen-rich gas. Adding a relatively modest amount of such gas to the gasoline powering a car or to a diesel vehicle's exhaust is known to have benefits for cutting the emissions of pollutants. "Prior to the Plasmatron reformer development, there was no attractive way to produce that hydrogen on board," stated by MIT's Cohn. The team is finding that the device could make vehicles cleaner and more efficient, with a potentially significant impact on oil consumption. The company ArvinMeritor has partnered with MIT to bring the Plasmatron to market and is currently being deployed and tested on city buses in Boston [4]. Ceramtec a company specializing in advanced ceramics material technology research and development focusing primarily in energy & environment (cleantech) areas has also developed a reformer using plasma technology. The Ceramtec plasma technology is an innovative, integrated process that exploits the thermodynamic advantages of steam reforming and combines it with the robust, highly energetic reaction zone characteristics of partial oxidation [1]. The ceramtec design can be seen in Fig 3.

NCSU has partnered with several corporations to develop a recycle swine production plant that has no waste. Steam reforming gasification is a form of thermal decomposition in an environment with limited or no oxygen. The concept is that material is indirectly heated to very high temperatures, for example, 800°C, at which point organic material decomposes into gases such as H₂, CO, CO₂, and CH₄, and ash containing minerals. A benefit of this process is that any bio-active compound, such as antibiotics, prions, or viruses, should be destroyed. The steam reforming gasifier used for research at NCSU uses an entrained flow principle. The swine feces are injected into a spiral tube while suspended by superheated steam and some recycled product gas. This tube surrounds an intense flame, and while traveling up this tube the material is heated to 800°C and decomposes. Steam is injected so it can react with fecal material resulting in H₂ production as seen in the Fig 4. In practicality, other product gases such as CO₂, CH₄, H₂S, and

NH₃ are formed as well. The product gas has a combustion value similar to low grade natural gas and can be used to fuel a generator or micro-turbine for the production of electricity. This is only a viable option if a market is available for electricity. The steam reforming gasifier does not require external energy for its operation. Instead, a portion of the product gas or, in the case of catalytic conversion of product gas to liquid fuels, unreacted product gas, is used to fuel the burner that generates the heat to sustain the process. Thus, the entire process is self-sustaining and equally important, the process does not produce any air emission of noxious compounds such as dioxins [7]. Large scale reformation is constantly being researched and applied by industrial companies, leaving the refining of the technology on a small scale to Universities.

Methods

Design

The basic design of the on-board fuel reformer was based upon many previously built devices used to assist in the operation small internal combustion engines. The schematic and 3-D model view of the final design created by the PEGASUS team are found in Fig 8 and Fig 9. A generic model of a reformer that was used as a foundation of the groups design is found in Fig 5.

Along with these designs, the specifications and limitations necessary to operate a fuel reformer on a push lawn mower contributed to the design that was chosen and assembled. The overall layout of the on-board fuel reformer was designed to accomplish only a few general tasks; these most basic tasks were to connect to the engine exhaust, heat and reform the fuel, then feed into the engine intake.

To accomplish these tasks, an array of piping and fittings were necessary to select and purchase as seen in Fig 6. Prior to assembly, several off the shelf pieces were necessary to machine to the appropriate lengths and diameters. Additionally, some soldering was necessary for the copper pieces (specifically for the bubbler exhaust intake). However, no pieces were welded in place. This decision was made to allow the reformer to be much more easily disassembled for testing and modification.

To connect to the engine exhaust, the stock muffler was removed and a pipe fitting was threaded in its place. This fitting leads to a 3 way connection that will take the exhaust into a 1" black pipe around a 1/2" black pipe containing the incoming fuel. This process essentially created a counterflow heat exchanger. This heat exchanger therefore allowed the heat from the exhaust gases to be transferred to the incoming fuel, thereby creating a syngas.

Upon exiting the heat exchanger portion of the reformer, another series of fitting and piping allowed the gas multiple flow paths. One path allows the gas exit through a muffler, much like a normal internal combustion engine. A secondary flow path was created by a T-junction with a ball valve controlling the percentage of flow. The second path directed the exhaust into a copper pipe leading into the bubbler. containing the alternate fuel. The still hot exhaust gases then

mixed with the fuel and gasified it in this container.

From the bubbler, the gasified fuel flowed through a second 3 way junction into the heat exchanger. This fuel, upon reformation, was pulled into the engine intake. To control the fuel/air mixture entering the engine manifold, a ball valve was used to regulate the incoming air. For the intake, the carburetor supplied by the engine manufacturer was removed. Instead, a pipe fitting was epoxied directly into the intake. This was an alteration to the original design, however. The group had expected a threaded connection to exist, similar to the exhaust.

The regulation of incoming air was a major component in the design and operation of the engine and reformer. As previously stated, without a carburetor, the engine had to be manually regulated. By controlling the ratio of fuel to air manually, the operator was able to directly affect the combustion reaction and the reformation process. The goal while operating the engine was therefore to determine a mixture that would allow operation of the engine while also allowing a temperature high enough to reform the fuel while also not causing damage to the engine or reformer.

A basic push lawn mower containing a Briggs and Stratton 4.5 ft-lbf, 4-cycle engine acted as the platform for the onboard fuel reformer. The reformer mounted directly to the lawn mower using a standard pipe hanger. The bubble was not mounted directly to the mower but was connected only by the incoming and outgoing hoses.

Assembly

Fig 8 is a schematic of the reformer assembly to follow as well as Fig 9 provides a 3-D model. The complete parts list to complete the proceeding assembly is found in Fig 6. A photograph of the final completed reformer can be seen in Fig 14.

1. Material Preparation

- a. Bore out one $\frac{1}{2}$ " end of the both the 1" x $\frac{1}{2}$ " x $\frac{1}{2}$ " reducing tee to $\frac{7}{8}$ "
- b. Face the bored out end of each of the reducing tees
- c. Cut $\frac{1}{4}$ " steel rod to 16 $\frac{7}{16}$ " in length
- d. Cut $\frac{1}{2}$ " copper pipe to length required to reach height of bubbler
- e. Solder a $\frac{1}{2}$ " to $\frac{1}{4}$ " NPT adapter to one end of copper pipe.
- f. Install cap on other end of copper pipe and drill a $\frac{1}{16}$ " hole
- g. Drill two $\frac{1}{2}$ " hole in top of bubbler
- h. Cut remaining copper pipe to length required to reach from the reformer to intake.

Install $\frac{1}{2}$ " Brass male flare fittings and flare nuts. Flare both ends.

- i. Cut three 3" x $\frac{1}{4}$ " nipples in half
- j. Drill $\frac{1}{4}$ " hole in lawnmower body to accept pipe hanger
- k. Connect $\frac{1}{2}$ " to $\frac{5}{8}$ " coupling to engine intake. May require epoxy or welding.

2. Engine Requirements and modifications

- a. Obtain small internal combustion engine with 4 cycle operation. Design must allow for adequate room to mount modified fuel/air intake and exhaust piping following modifications.

- b. Remove engine gas tank, muffler and carburetor

3. Reformer Assembly

- a. Connect prepared 1" x 1/2" x 1/2" reducing tees to both ends of 1/2" black pipe. Tighten fully.
- b. Slide 1/2" black pipe into 1" pipe (may require some force). Ensure 1/2" pipe is fitted evenly with equal exposed threads on each side.
- c. Connect 1/2" pipe tee to 1/2" to exposed portion of black pipe on bubbler side of the reformer. Use 1/2" copper washers in between fittings. Tighten down enough to crush washer and form seal.
- d. Connect 1/2" x 1/2" coupling to exposed portion of 1/2" black pipe on the exhaust side of reformer. Use copper washer and tighten similarly to bubbler side.
- e. Ensure fittings on body of reformer are tightened sufficiently. This may require each fitting to be adjusted individually to align each section.

4. Exhaust Side of Reformer

- a. Connect 1/2" x 1/2" x 1/2" tee to 1/2" coupling using 1/2" nipple.
- b. Connect 1/2" NPT ball valve to inward facing connection of tee using 1/2" nipple.
- c. Connect additional 1/2" close nipple to other side of ball valve
- d. Attach 1/2" x 1/2" 90 Degree elbow to nipple. Turn to face along length of reformer.
- e. On longitudinal facing connection of tee, connect 1/2" x 5/8" reducing coupling using 1/2" close nipple.
- f. Connect 5/8" x 1/2" copper reducing coupling to end of reformer section.

5. Bubbler Side of Reformer

- a. On bubbler side of reformer, connect 1/2" close nipple to 3-way tee facing upward. This will be for piping connecting to muffler.
- b. Connect a 1/2" x 1/2" x 1/2" tee to the upward facing nipple
- c. On the inward facing connection of the tee, connect a 1/2" close nipple.
- d. Attached a 1/2" to 1/4" reducing coupling to nipple.

- e. Attach cut 3'' x 1/4'' pipe nipple to reducing tee.
- f. The second 1/2'' NPT ball valve is connected to the top of the 1/2'' x 1/2'' x 1/2'' tee.
- g. The original muffler, or newly purchased muffler, is this connected to the ball valve using a 1/2'' close nipple.
- h. Using the exposed portion of the 1/2'' black pipe, connect the middle fitting of a 1/2'' x 1/2'' x 1/2'' tee.
- i. On each side of the tee, connect a 1/2'' close nipple followed by a 1/2'' ball valve.
- j. On the side of the tee facing the engine, connect a 1/2'' close nipple followed by a 1/2'' by 1/4'' reducing coupling.
- k. Connect half of the cut 3'' x 1/4'' nipple to the reducing tee.

6. Attaching reformer to bubbler

- a. Take copper pipe with soldered 1/2'' to 1/4'' NPT adapter and connect 1/2'' x 1/4'' street elbow (male fitting will thread into copper adapter).
- b. On remaining side of elbow, connect cut 3'' x 1/4'' pipe nipple.
- c. Connect 1/2'' ID reinforced vinyl hose to cut nipple. Clamp down with 1/2'' hose clamps.
- d. Connect other end of 1/2'' ID reinforced vinyl hose to cut 3'' x 1/4'' pipe nipple below the muffler on the reformer. Clamp down with hose clamps.
- e. Push copper pipe (with connection and hose) into top of bubbler. Seal penetration point.
- f. Connect second length of vinyl hose to cut 3'' x 1/4'' pipe nipple on the lower section of the bubbler side of the reformer. Clamp down using hose clamps.
- g. Connect other end of hose to cut 3'' x 1/4'' pipe nipple.
- h. The other end of this nipple is threaded into the copper adapter.
- i. Push adapter into second penetration in top of bubbler and seal.

7. Attaching Reformer to Engine Intake and Exhaust

- a. Thread 2'' x 1/2'' pipe nipple completely into exhaust manifold of engine.
- b. Place the exhaust intake of reformer (1' x 1/2'' x 1/2'' tee) up to the exhaust manifold pipe nipple.

- c. Back the nipple out of the exhaust manifold while threading into the reformer intake tee (necessary due to both having right handed threads. Process will leave each side about halfway threaded).
- d. Connect pipe hanger to lawnmower body using washers and lock nuts on each side of the sheet metal.
- e. Use pipe hanger to support back side of reformer (front side is supported by exhaust manifold nipple).
- f. Bend flared copper pipe to radius required to connect reformer to intake manifold.
- g. On end of exhaust side of reformer, connect copper pipe fittings to coupling.
- h. Loop copper pipe around front on engine and connect to intake coupling.
- i. On 90 Degree elbow, connect original gasoline tank using a 1/2" coupling.

Operation

Before the fuel reformation was able to begin, a warm up time was needed to get the reactor to temperature and the plasma present. This period did not last long because as soon as combustion began, heat was immediately sent through the exhaust. The initial plan was to use gasoline to warm the system up, and then chose from a wide variety of fuels to add to the system in virtually any concentration. Alcohol was chosen for the alternate fuel due to its widespread availability, and in the rare event of it being unavailable commercially, it can be made by anyone at home with a few basic ingredients that are commonly found in households. Being that alcohol burns a little cooler than gasoline, plans are to run the engine slightly lean to reach a higher temperature in the reactor. The following descriptions of operation all refer to Figure 5. Once everything is warmed up, fuel reformation may now begin. Exhaust leaves the engine and enters the piping labeled exhaust in the top left of the diagram, in which it then enters the reactor and travels through the area shaded blue. Heat is transferred to the pipe containing fuel and the steel rod suspended within, and then exits the chamber on the opposite end and has the choice of either exiting through the muffler or recycled further into preheating and aerating the fuel mixture. The air/fuel mixture then leaves the bubbler (volatilization chamber) via vacuum through the hose located at the top left of the bubbler, and enters the reactor travelling through the area shaded with a peach color. With the outer section of the reaction chamber being heated by the engine exhaust, the cooled output from the volatilization chamber makes its way through the inner section of the reaction chamber, passing between the outer wall and a magnetized steel rod placed in at the center. This inner section of the reaction chamber imposes a magnetic field upon the molecules leaving the volatilization chamber, magnetically polarizing the hydrogen, molecular oxygen and carbon monoxide molecules. The result of this process is the re-formation of molecules. While the steam and vaporized fuel combine in a steam reformation process resulting in syngas + hydrogen, the plasma further promotes more hydrogen generation. The hydrogen rich syngas then exits the left side of the reactor exiting through the piping labeled intake and enters the engine, for a clean and efficient burning of the new fuel that was created. After combustion, the process is repeated.

While all of the components of the entire system are required for operation, the most vital parts of the system are the bubbler and the reaction chamber. In the bubbler, or the volatilization chamber, is the first place that the alternate fuel sees before going anywhere else. The bubbler consists of two components, a high pressure/high temperature input which is bubbled through the alternate fuel and a low pressure output located in the top of the container for vapors to escape through to begin their journey to the reaction chamber. Through high pressure and high temperature, the alternate fuels begin to vaporize in the bubbler and then exit through the low pressure output. The next place that these vapors will enter, after the addition of atmospheric air through the use of a valve, is the reaction chamber. The purpose of the reaction chamber is not only to reform the fuel, but also to add energy to the fuel and to ensure that all fuels entering have reached their flashpoint (through heat exchange), which is the temperature at which a liquid will react with air and combust when a flame is present. Upon leaving this chamber is the point at which the new fuel enters the engine for combustion.

Operating the engine with this system attached is fairly straight forward, and it gives the operator total control over air/fuel ratios and engine RPM. There are 2 valves, one at the muffler and one right before the reactor, that control the air and back pressure. The back pressure and air valves are dependent on each other, and have to be adjusted together. Once these settings are locked in, the motor is ready to run until the fuel runs out. The other valve, the one directly past the output of the bubbler, controls the amount of fuel and can be directly used to control the RPM of the enging. While all three valves are used together, the only 2 that have to be adjusted at the same time are the air and back pressure valves to ensure correct pressures in the bubbler.

Results

The results to the project are found in several different forms. As the project goals were to design, build and explain the operation of an internal combustion engine using a petroleum/gasoline substitute system, the attached figures are evidence that those processes were attempted and completed. The final figures include the photos of the completed reformer and fuels used to operate the engine. The goals of the project, however, were not to obtain specific data on the operation and efficiency of the reformer and engine. Those goals will be completed by a separated testing team. The lawnmower was first started and ran using the valve connecting the gasoline tank to the engine with the reformer closed off from the system. The engine started and ran using only the normal function of the gasoline tank. The next result achieved was when the gasoline tank valve was closed and gasoline added to the bubbler. The reformer system was then opened and the lawnmower started and ran on only gasoline supplied into the bubbler. With the reformer in proven working order and all components functioning correctly the next results obtained were the starting and running of the engine through the reformer with a variety of fuels. E85 was used as the sole fuel in the bubbler, and the engine started and ran. Next methanol, pure ethyl alcohol, and diesel were all tried separately to start and run the engine. The only fuel tried that couldn't be used to solely start and run the engine was the diesel. A mixture of 1 part E85 to 3 parts diesel was ran through the system and successfully started and ran the engine. Additionally, a 50/50 mixture of methanol and E85 was used and ran the motor with success. A 50/50 mix pure alcohol and methanol started and ran the engine. Finally a mix of $\frac{1}{3}$ Diesel, $\frac{1}{3}$ Methanol, and $\frac{1}{3}$ E85 was used resulting in the engine starting and running. Although the efficiency of each fuel could not be tested, previous studies have determined the possible efficiencies of these fuels based upon their molecular makeup. For the gasoline, methanol, ethanol, and E85, efficiency increases respectively were 20%, 13.88%, 16.67%, and 28.57% as can also be seen in Fig 11 and Fig 12.

Discussion

In order to assist in the understanding and development of a petroleum/gasoline substitute system, the design group undertook the task of designing, building, describing the operation, and if time allowed, testing of an onboard fuel reformer. The reformer was designed per requirements set by Dr. Knisley and research done on similar systems. Per the initial requirements, the onboard fuel reformer was constructed in a manner to mount to an internal

combustion engine on a lawnmower. The design allowed for an array of alternate fuels or gasoline/alternate fuel mixtures to be tested. The design also allowed for the fuel/air ratio to be easily varied as well as modification and additions to be made. The device was operated and this operation can be fully explained and verified.

The central requirement of the PEGASUS project was to design and physically construct a petroleum/gasoline substitute system and was completed by using a system known as an onboard fuel reformer. The photographs located in Fig 14 through Fig 16 are show that this was accomplished. The accompanying videos on the group website located at the project website at dasp.mem.odu.edu:8080/~pegasus_fa12/index.htm show that this system was functional and was able to be operated and adjusted. Because these project guidelines were to construct the device, there is little numerical data obtained to interpret and compare. What can be conclude from the accomplishment of the fundamental project goal is that the system designed by the group could be easily and affordable completed. This system was also functional in that it allowed operation of the engine to be adjusted by controlling the air intake, exhaust outlet, and fuel intake ball valves thereby giving the operator full control of air/fuel ratios and engine RPM. The results also show that both non-petroleum fuels and unconventional petroleum mixtures such as ethanol, methanol, ethanol/diesel mixtures and other combinations were able to run the engine.

The results from operating the engine on multiple fuels after they are reformed can be interpreted several ways. The result of the engine running on only gasoline first through the gas tank and then through the reformer system validates our design for testing. Using the operation with gasoline in the gasoline tank gave us a baseline to compare operation through the reformer system. Since our lawn mower was not capable of having ethanol, methanol, or E85 run through the motor via the starting fuel container without permanent modifications, baselines of these fuels in comparison to gasoline were taken from an online source in which the table can be seen in Fig 11 and Fig 12. One result gained was just solely the fact that the engine was able to run on various fuels other than gasoline, both petroleum and non-petroleum liquids. From the factory, the engine is calibrated for gasoline only, and if any other type of fuel is poured in

the gas tank, the motor is either to not run or it will fail shortly after running if it does start. We successfully ran diesel, ethanol, methanol, and E85 in addition to gasoline through the system; as a result goal number one was a success. With the previous baselines as mentioned earlier along with data gathered from running various fuels through the engine, efficiency increases were able to be estimated. While the data gathered this semester was limited due to unforeseen issues with the build and time constraints, the information gathered was conclusive and also validated the operation of the reformer.

The limitations outside the obvious constraints of budget the project encountered were materials and testing ability. Since this semesters group focused on the design and build portion, materials and parts available locally for purchase were the major limitations encountered. The layout of the lawnmower as well as the needed reactor space put limitations as to the design of the reformer system itself. The group was unable to find a proper hose that was transparent and could also withstand the temperatures produced at the exhaust. The reformer could only be run for 20-30 minutes before the hose would melt. Having to change out the hose allowed the reformer to cool and not reach extreme temperature values. The sealants used also would break down after extensive testing as result of heat in combination with corrosive fuels. Without proper seals it was hard to determine the required air/fuel ratio to maintain the same RPM as other trials in order to gather accurate data. Solving this issue permanently would requiring finding a sealant capable of withstanding both high temperatures and corrosive fuels. Another limitation encountered would be the measuring of the reactor temperature. A temperature reading can be achieved using a thermo coupling, or thermometer attached to the reactor that is well insulated but designing an internal temperature measuring device would cause sealant issues allowing the fuel and exhaust to mix. Even with these limitations, valid results were obtained as to the design and operation of the reformer. The implication of these results are that the current design can be used to operate a lawn mower through a reforming process on a variety of fuels and fuel mixtures. This reformer can be used in emergency situations when gasoline is scarce or contaminated. This system has proven its ability to work, and with further testing will provide insight as to alternative fuel sources and mixes that can be used to improve performance and efficiency on small engine operation. This system provides an alternative to

gasoline fueled engine operation, with the ability to run on E85, methanol, alcohol, a mix of the prior listed with diesel, and many other non-petroleum liquids as long as the flash point is under 600 degrees fahrenheit. The efficiencies increases respectively were 20%, 13.88%, 16.67%, and 28.57% as can also be seen in Fig 11 and Fig 12. This process, after fully understood, could be scaled up to operate on transportation vehicles.

Conclusion:

Several important conclusions can be drawn from the results of the PEGASUS design project. Additionally, there is great potential for additional research and conclusions with more involved testing. With a completed operating reformer prototype, future testing by a second group will provide data to further analyze for efficiency and power. By using a temperature gauge, tachometer, and dynamometer, a great amount of data could be collected to determine the engine's exact operating characteristics. From the research phase of the project, it became clear that reformation technology is quite developed, yet only a small portion of this technology has been utilized to increase fuel efficiency of personal vehicles and even less for that of small internal combustion engines. The design phase utilized the information gathered during the opening research weeks of the project. The design necessary to create a reformer was very simple in concept yet included all of the elements necessary for exhaust gas reformation. Similarly, the construction of the reformer was relatively simple and required only components that could be purchased at hardware stores. Simply put, the preceding report shows that an exhaust gas reformer can be easily and inexpensively added to a small internal combustion engine. A more explicit conclusion can be drawn through the actual operation of the reformer. Using the constructed device, the engine was able to operate on various alcohols and mixtures including diesel, something that would not be possible with a normal spark engine. These results show that the reformer was indeed functioning in the intended manner. Therefore, by installing the reformer, an internal combustion engine was given the ability to operate using a multitude of combustible substances.

The ability of an engine to run on non-petroleum fuels and unorthodox fuels is very significant. There are numerous reasons why the sole reliance on petroleum based fuels is unwise and unsustainable. The capability of allowing existing engines to expand their potential fuels

sources is a small yet significant step in allowing a departure from this reliance. Although more testing is necessary to verify any increases in efficiency and reduction in emission, the onboard fuel reformer has shown great potential in moving petroleum/gasoline systems forward.

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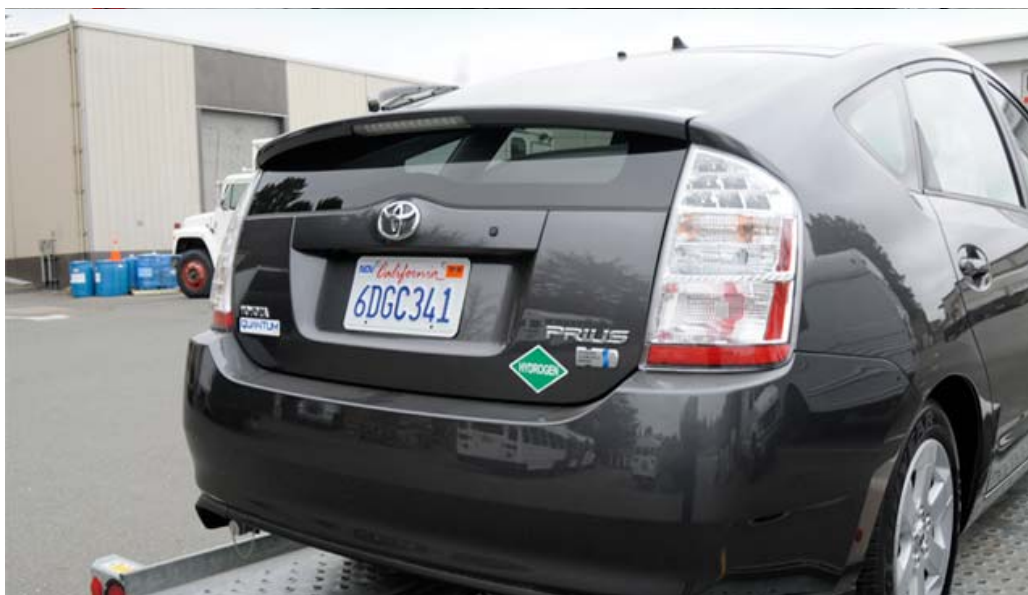
Appendices

Figure 1



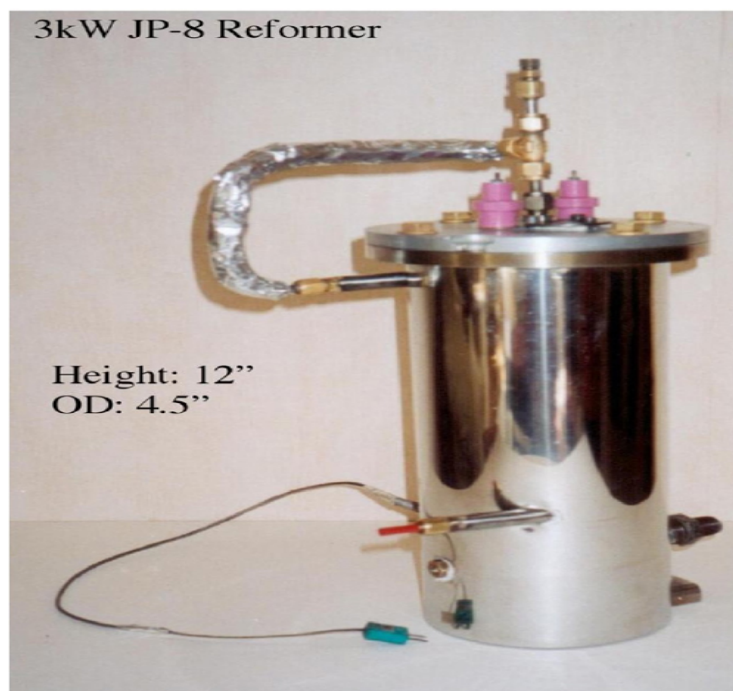
Topsoe ATR Reformer Facility

Figure 2



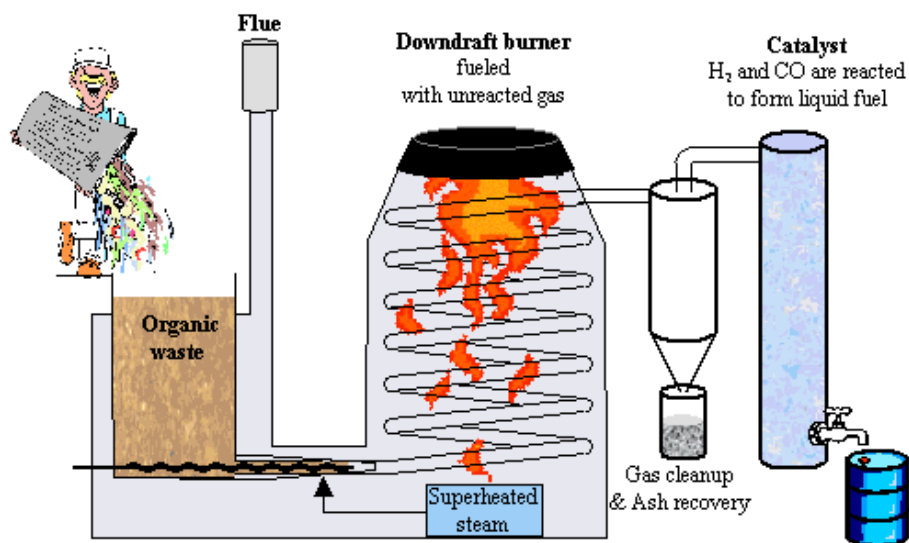
Toyota Hydrogen Initiative

Figure 3



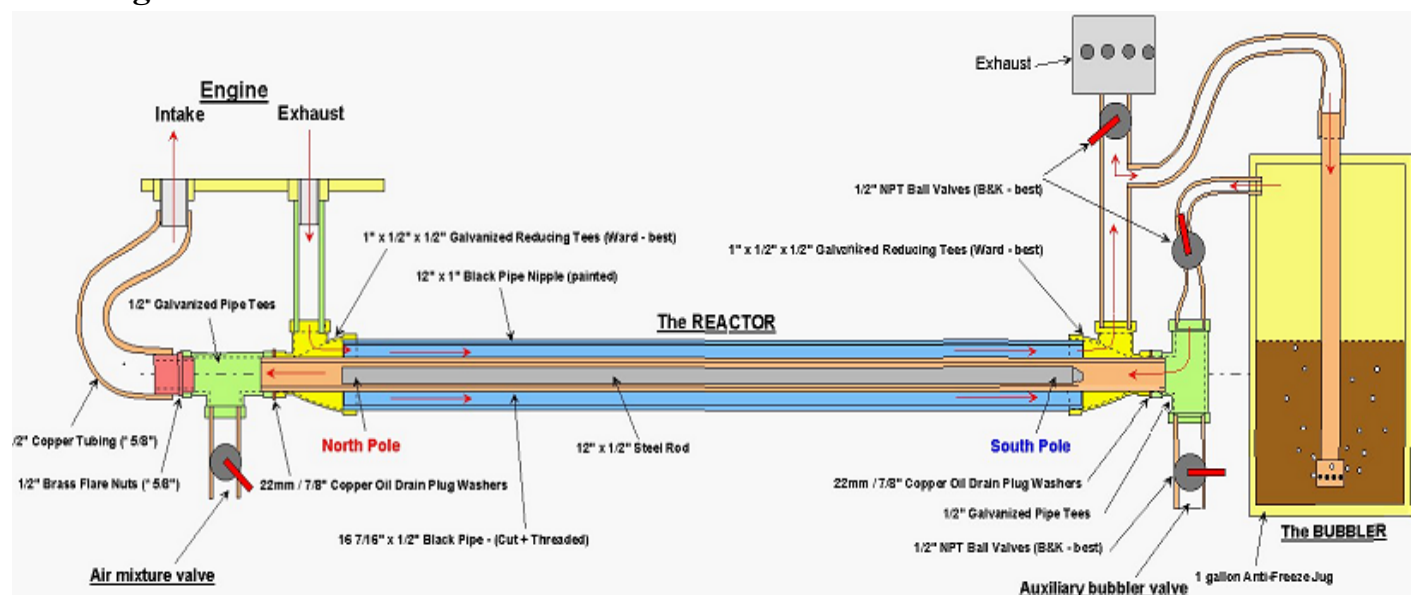
Ceramatec Fuel Reformer

Figure 4



Recycle Swine Production by NCSU

Figure 5



On-board Fuel Reformer Design

Figure 6

Required Materials

Part		Quantity
1.	16 7/16 x 1/2" Black Pipe (Cut and Threaded)	1
2.	12" x 1" Black Pipe	1
3.	12" x 1/4" Steel Rod	1
4.	1" x 1/2" x 1/2" Reducing Tee	2
5.	7/8" Copper Oil Drain Plug	2
6.	2" x 1/2" Galvanized Pipe Nipple	1
7.	1" Galvanized Pipe Hanger	1
8.	1/2" Close Pipe Nipples	8
9.	1/2" NPT Ball Valves	4
10.	1/2" Galvanized NPT Muffler	1
11.	1/2" Galvanized Pipe Tees	4
12.	1/2" x 1/4" Galvanized Pipe Reducers	2
13.	High Temperature Grill Paint	1
14.	6' x 1/2" ID Reinforce Clear Vinyl Hose	1
15.	3" x 1/4" Galvanized Pipe Nipple	3
16.	9/16" Copper Washers	2
17.	1/2" Copper Tubing	1
18.	1/2" to 5/8" NPT Brass Male Flare Fitting	2
19.	1/2" Brass Flare Nuts	2
20.	1 Gallon Plastic Container	1
21.	1/2" Hose Clamps	4
22.	1/4" Galvanized Street Pipe Elbow	1
23.	1/2" Copper Pipe Cap	1
24.	3" x 1/2" Galvanized Pipe Nipple	1
25.	Solder	1

Figure 7

Table 1.1

<i>Steam reforming</i>	<i>Combustion</i>
Endothermic reaction which consumes heat	exothermic reaction which generates heat
Methanol + steam => hydrogen + carbon dioxide $\text{CH}_3\text{OH} + \text{H}_2\text{O} \Rightarrow 3 \text{H}_2 + \text{CO}_2$	methanol + air => carbon dioxide + water $2 \text{CH}_3\text{OH} + 3 \text{O}_2 \Rightarrow 2 \text{CO}_2 + 4 \text{H}_2\text{O}$ hydrogen + air => water $2 \text{H}_2 + \text{O}_2 \Rightarrow 2 \text{H}_2\text{O}$

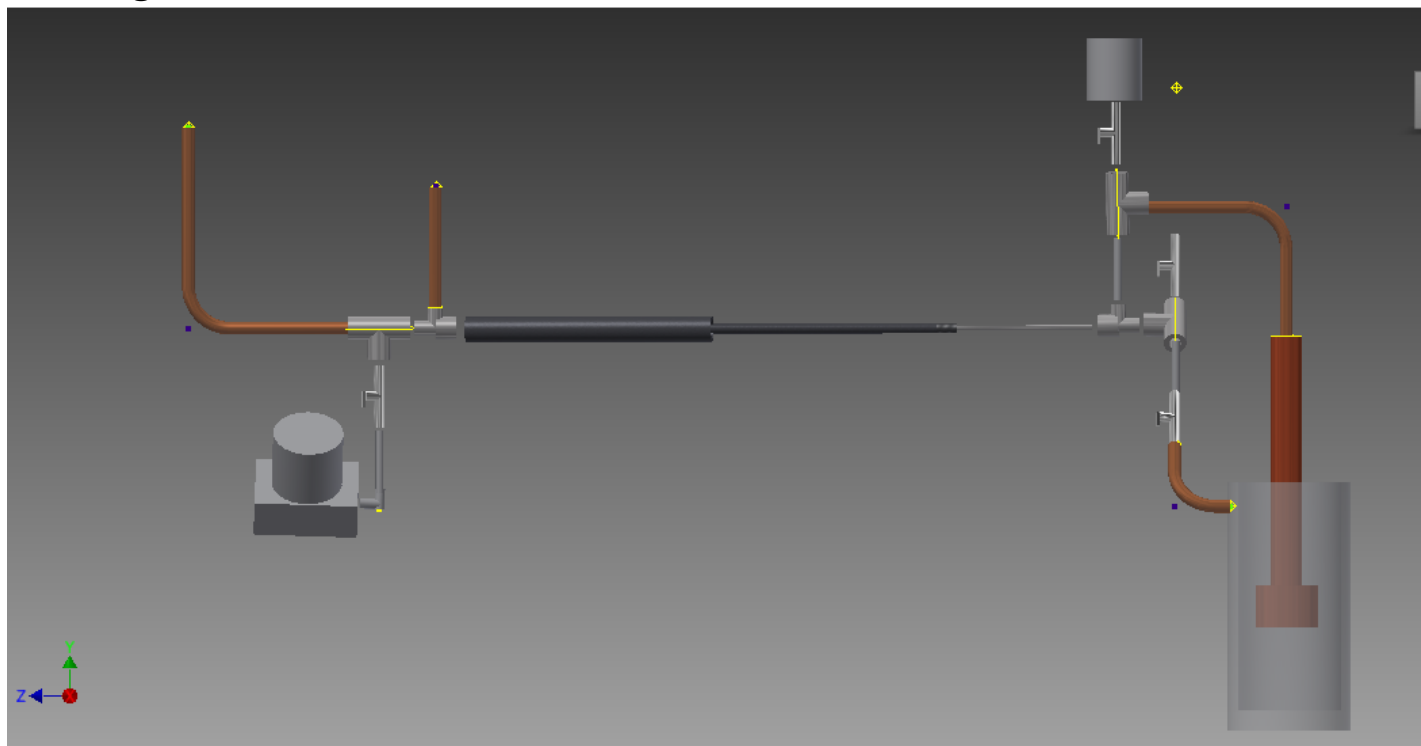
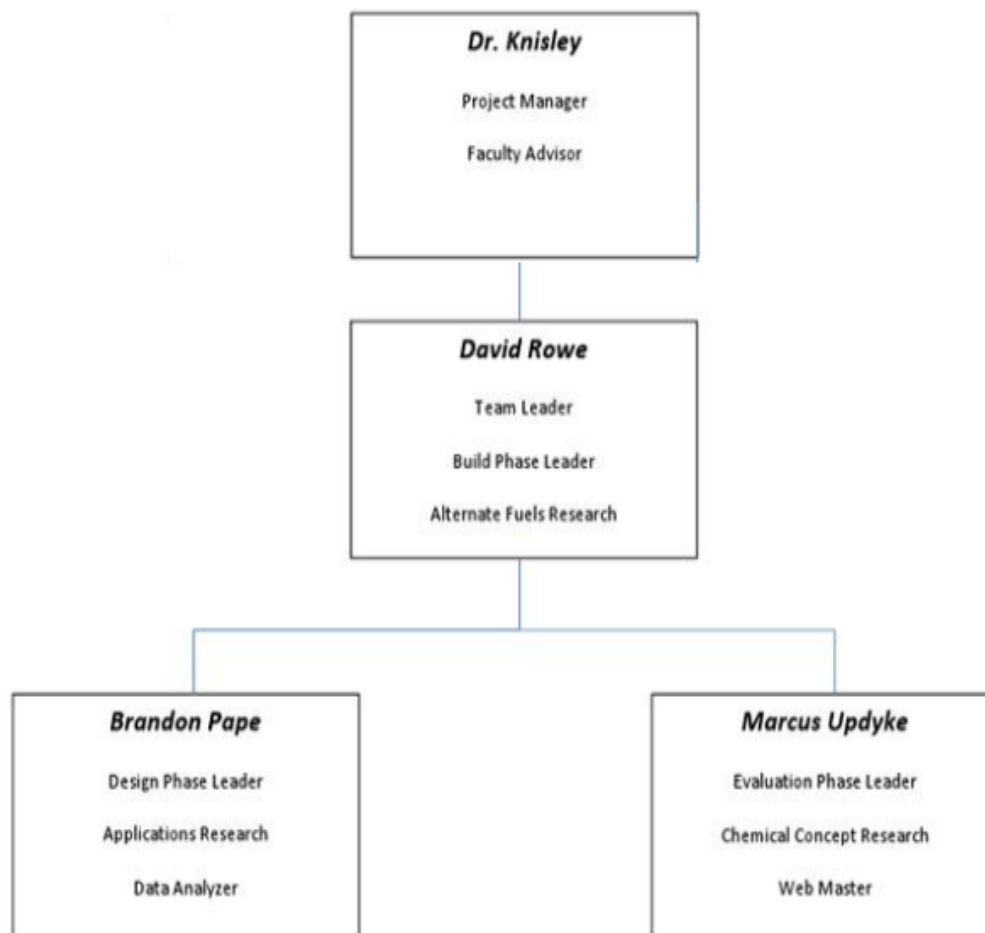
Figure 8

Figure 10**Figure 11**

Liquid Fuel	Equivelant Tank of Gasoline w/o System	Equivalent Tank of Gasoline w/ System	Efficiency Increase
Gasoline	1	0.8	20%
Ethanol	1.5	1.25	16.67%
Methanol	1.8	1.55	13.88%
E85	1.4	1	28.57%
Diesel	N/A	Will now operate a gasoline engine	N/A

Figure 12

Property	Methanol	Ethanol	Gasoline (87 Octane)	E85
Chemical formula	CH ₃ OH	C ₂ H ₅ OH	C ₄ to C ₁₂ chains	*
Main constituents (% by weight)	38 C, 12 H, 50 O	52 C, 13 H, 35 O	85–88 C, 12–15 H	57 C, 13 H, 30 O
Octane (R+M)/2	100	98-100	86-94	105
Lower heating value (Btu/lb)	8,570	11,500	18,000-19,000	12,500
Gallon equivalent	1.8	1.5	1	1.4
Miles per gallon compared to gasoline	55%	70%	–	72%
Relative tank size to yield (driving range equivalent to gasoline)	Tank is 1.8 times larger	Tank is 1.5 times larger	1	Tank is 1.4 times larger
Reid vapor pressure (psi)	4.6	2.3	8–15	6–12
Ignition point - Fuel in air (%)	7–36	3–19	1–8	*
Temperature (approx.) (°F)	800	850	495	*
Specific gravity (60/65°F)	0.796	0.794	0.72–0.78	0.78
Cold weather starting	Poor	Poor	Good	Good
Vehicle power	4% power increase	5% power increase	standard	3%–5% power increase
Stoichiometric air/fuel ratio (by weight)	6.45	9	14.7	10

* Depends on percentage and type of the hydrocarbon fraction.

Fig 13 Gantt Chart:

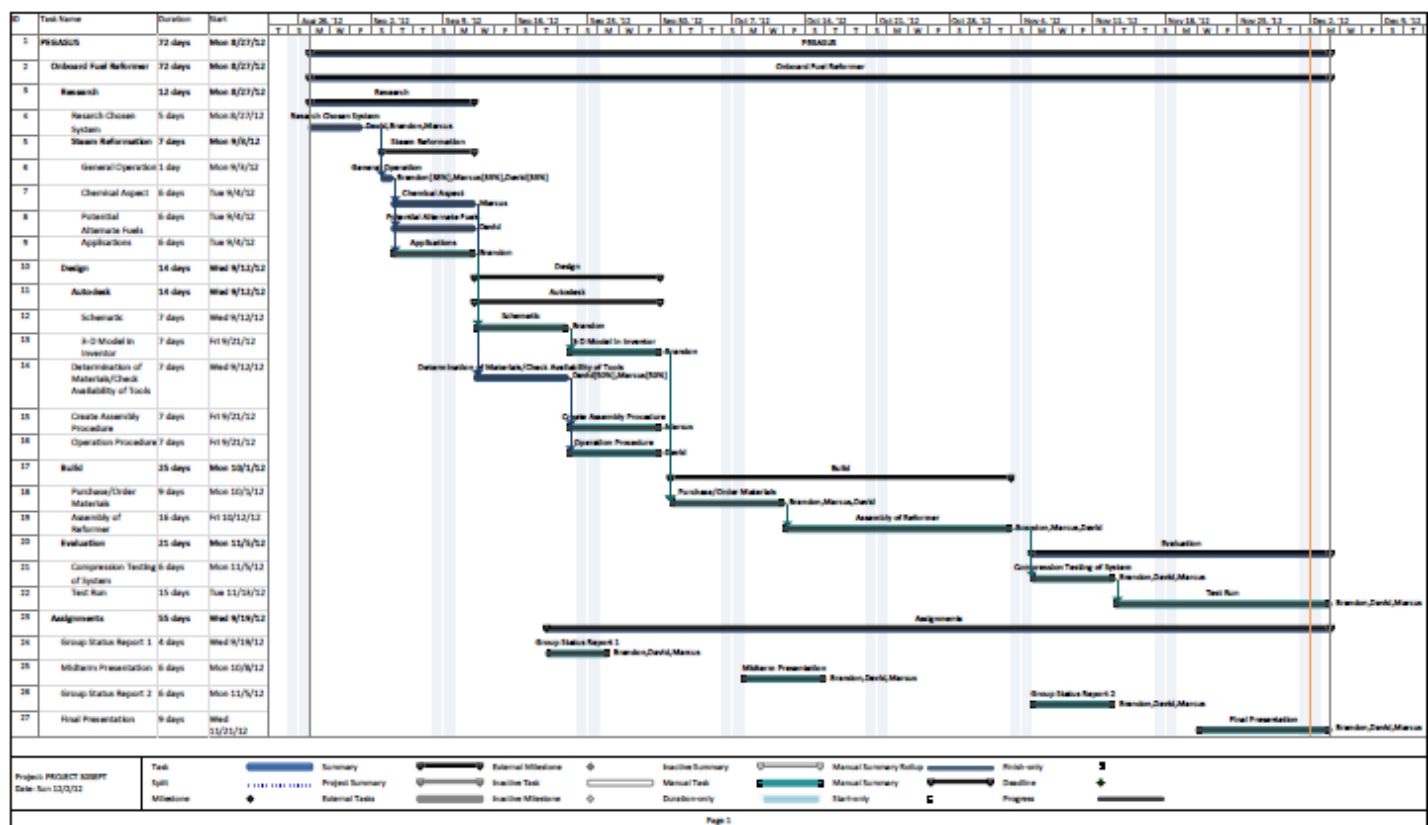


Fig 14
Completed Reformer mounted to engine



Fig 15
Completed reformer mounted to engine



Fig 16
Assembled reformer

